



Optimizing the Automotive Customer Journey **through Salesforce Automation and Geolocation**

ABSTRACT:

This project presents the implementation of a tailored Salesforce CRM solution for WhatNext Vision Motors, a pioneering automotive leader dedicated to transforming the mobility sector through customer-centric technology. The primary objective was to modernize the vehicle procurement journey by streamlining the customer ordering process and optimizing inventory management through advanced automation.

The solution features a geolocation-aware system designed to automatically assign the nearest dealership to customers based on their address, significantly enhancing convenience and reducing friction during the purchase phase. To maintain operational integrity, a robust inventory validation mechanism was established to prevent the placement of orders for out-of-stock vehicles, thereby ensuring data accuracy and proactive communication.

Furthermore, the project leverages Scheduled Flows and Apex to manage bulk order record updates. This automated logic dynamically adjusts order statuses to 'Confirmed' or 'Pending' based on real-time stock availability. By digitizing these critical touchpoints, the system eliminates manual administrative burdens, allowing staff to focus on strategic growth rather than routine data entry.

This end-to-end implementation provides WhatNext Vision Motors with a scalable, high-performance platform that ensures transparent order fulfilment, minimizes human error, and fosters long-term customer loyalty through a sophisticated and efficient digital experience.

OBJECTIVE:

The main objective of this project is to develop and implement a customized Salesforce CRM solution for WhatNext Vision Motors to streamline core business operations, maintain data integrity, and enhance customer satisfaction.

By building a centralized system to manage customers, vehicle orders, inventory, and dealer locations, the project aims to:

- Automate key processes such as dealer assignments based on geolocation and automated order status updates.
- Ensure accurate and consistent data entry by preventing orders for vehicles that are currently out of stock.
- Enable real-time visibility of vehicle availability and fulfilment status for both staff and customers.
- Improve operational efficiency by automating bulk record updates, allowing staff to focus on strategic, high-value tasks.
- Deliver personalized customer experiences through efficient ordering, reduced errors, and transparent communication regarding order confirmation.

TECHNOLOGY DESCRIPTION:

Salesforce

Salesforce is a cloud-based Customer Relationship Management (CRM) platform that helps businesses manage customer data, automate processes, and improve service, marketing, and sales operations. It provides point-and-click tools as well as programmatic capabilities (like Apex and Flows) to build custom business solutions.

Custom Objects

Objects in Salesforce are like tables in a database. Custom Objects are created to store specific business data.

Example:

- Vehicle__c – Stores details about car models and specifications.
- Order__c – Stores customer purchase records.
- Dealer__c – Stores information about showroom locations.

Tabs

Tabs are used to display object data in the Salesforce UI.

Example: A tab for Order__c allows dealership staff to easily view and manage pending vehicle deliveries.

Custom App

An App in Salesforce is a collection of tabs grouped together for a specific business purpose, such as a "Vehicle Management App" for the sales team.

Profiles

Define what a user can see and do (e.g., Sales Reps can create orders, but only Managers can delete them).

Roles

Control data visibility based on a hierarchy, ensuring regional managers can see data for all dealerships in their territory.

Validation Rules

Validation Rules ensure data entered meets business criteria.

Example: An order cannot be saved if the vehicle's Stock Quantity is zero.

Flows

Flows automate business logic without code.

Example: Record-Triggered Flow: Automatically finds the nearest dealer when a new order is created.

- Scheduled Flow: Runs nightly to check inventory and update bulk order statuses to "Confirmed" or "Pending".

Apex

Apex is Salesforce's object-oriented programming language used for complex custom logic.

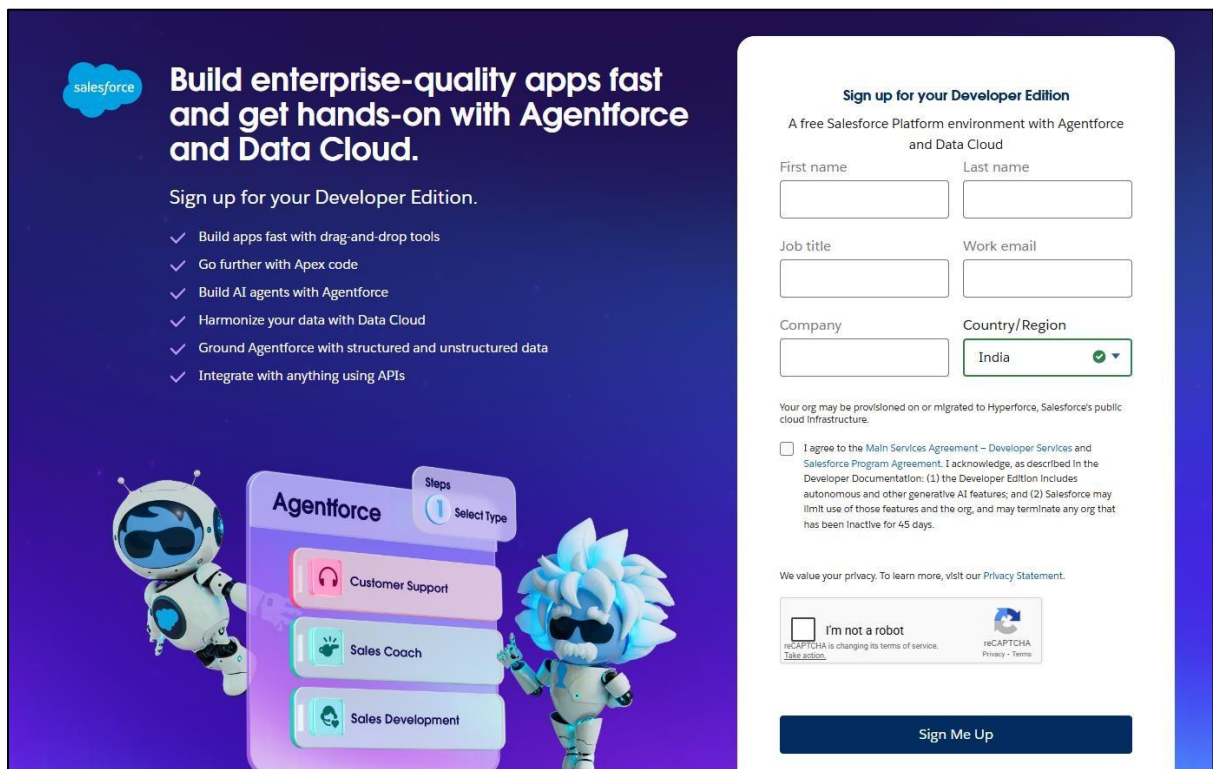
Example Trigger: Automatically recalculate regional tax and shipping fees on an order.

- Update real-time inventory counts across multiple dealership locations.

DETAILED EXECUTION OF PROJECT PHASE

1. Developer Setup Org:

- A Salesforce developer org was created using <https://developer.salesforce.com/signup>
- The account was verified, password set and access was granted to the Salesforce set up page.



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Steps
1 Select Type

- Customer Support
- Sales Coach
- Sales Development

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First name Last name

Job title Work email

Company Country/Region

India

Your org may be provisioned on or migrated to Hyperforce, Salesforce's public cloud Infrastructure.

☐ I agree to the Main Services Agreement - Developer Services and Salesforce Program Agreement. I acknowledge, as described in the Developer Documentation: (1) the Developer Edition Includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

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Privacy - Terms

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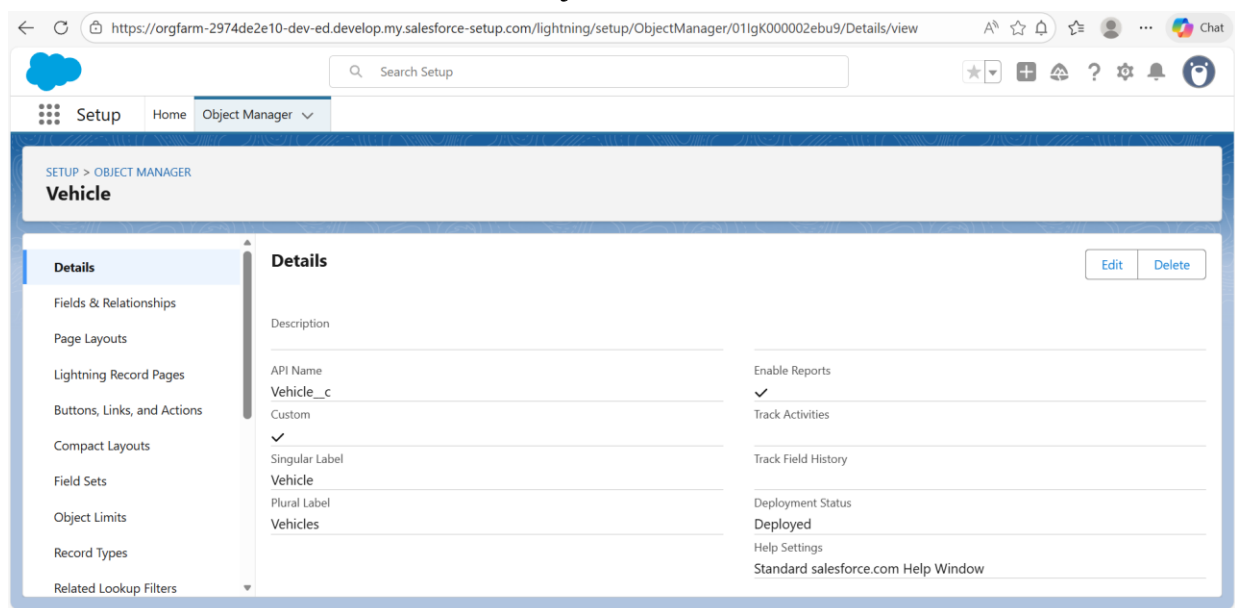
2. Custom Object Creation

Six custom objects were created to store business critical data:

Object Name	Purpose	Relationships
Vehicle__c	Stores vehicle details	Related to Dealer & Orders
Vehicle_Dealer__c	Stores authorized dealer info	Related to Orders
Vehicle_Customer__c	Stores customer details	Related to Orders & Test Drives
Vehicle_Order__c	Tracks vehicle purchases	Related to Customer & Vehicle
Vehicle_Test_Drive__c	Tracks test drive bookings	Related to Customer & Vehicle
Vehicle_Service_Request__c	Tracks vehicle servicing requests	Related to Customer & Vehicle

Steps Followed:

- Navigated to setup → Object manager → Create → Custom object
- Provided label, name and enabled reports/search
- Saved and created Tabs for each object



3. Creating the lightning app

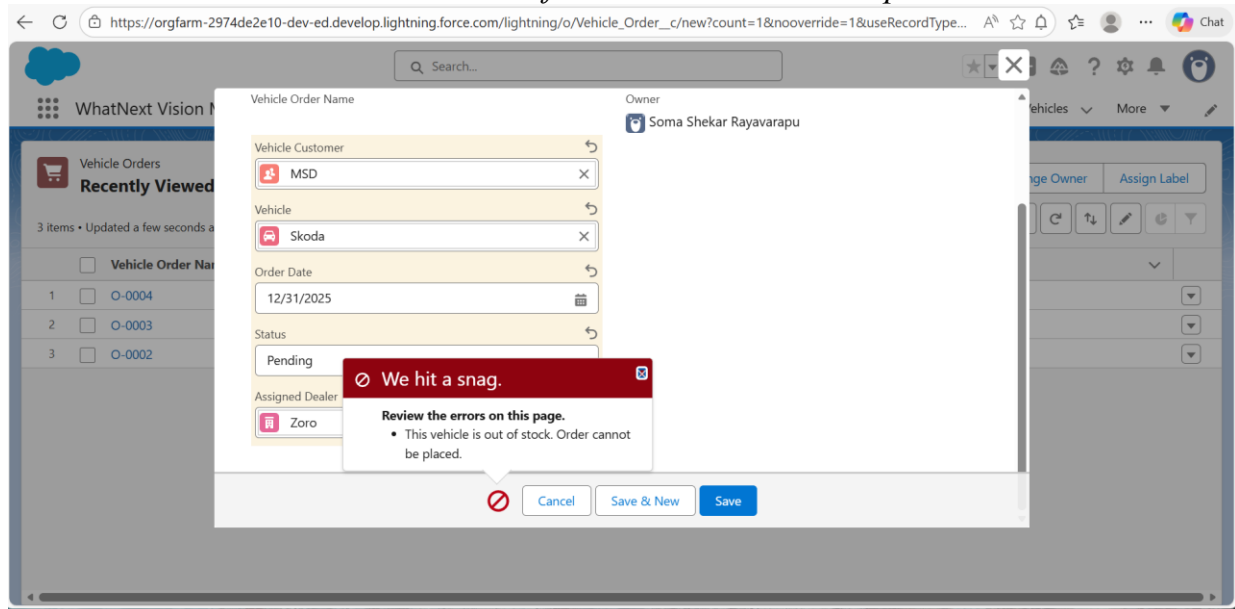
- A custom lightning App named **WhatNext Vision Motors** was created.
- Included tabs: Vehicle Customers, Vehicle Dealers, Vehicle Orders, Vehicle Test Drives, Vehicles, Reports, Dashboards, etc.,
- Assigned to System Administrator profile.

4. Validation Rules

To ensure accurate data entry and enforce business logic, the following validation rules were applied:

- Order Object: Prevents saving if Stock_Quantity<=0

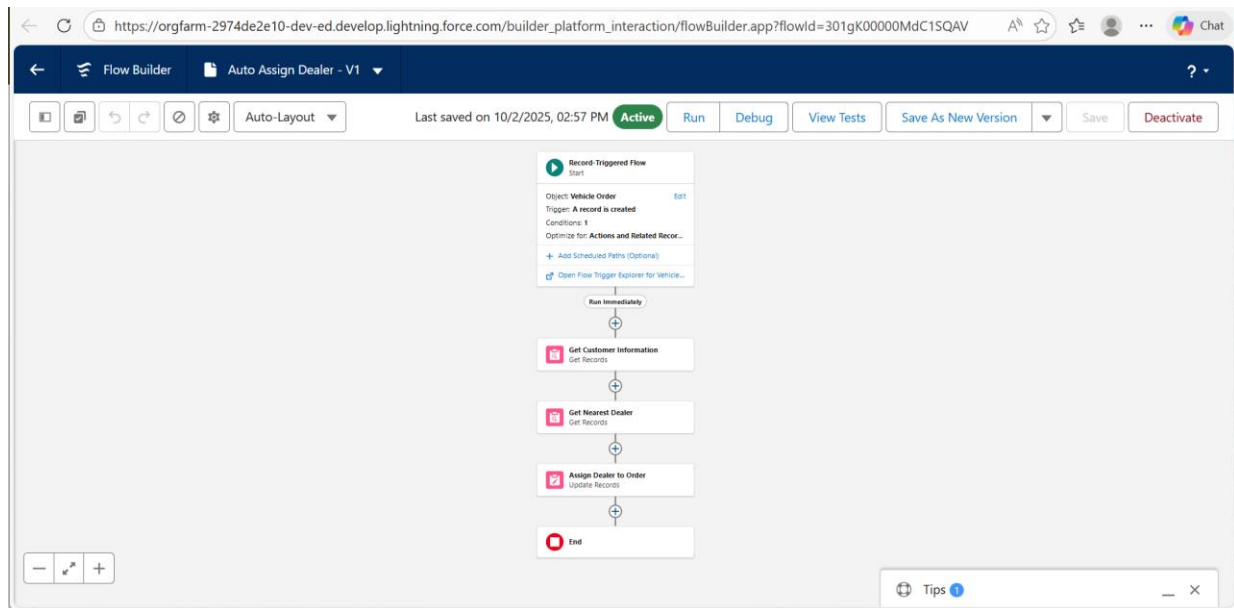
Error: "This vehicle is out of stock. Order cannot be placed."



5. Flows

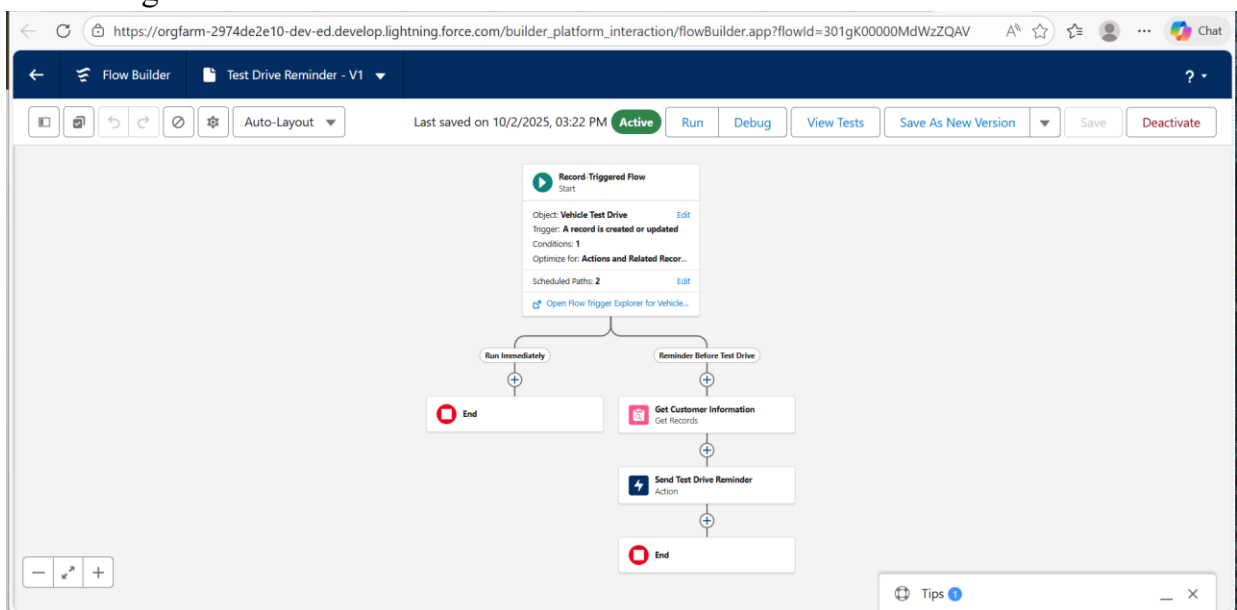
Automated Test Drive Notification Flow

- The Test Drive Reminder Notification flow is a vital customer engagement tool designed to reduce "no-shows" and maintain professional communication standards.
- This flow monitors the Vehicle Test Drive object for records marked with a 'Scheduled' status. By utilizing a Scheduled Path within the flow, the system can automatically trigger an email alert to the customer exactly 24 hours before their appointment date.
- This proactive approach ensures that customers are well-informed and provides them with the necessary details, such as the vehicle model and showroom location, directly in their inbox.



Test Drive Remainder Flow

- Enhancing Customer Experience through Timely Alerts By automating the communication loop, WhatsNext Vision Motors ensures a consistent and reliable experience for every potential buyer.
- The flow leverages the relationship between the Test Drive, Customer, and Vehicle objects to dynamically pull relevant data into a predefined email template.
- This automation not only saves significant manual effort for the dealership staff but also reinforces the brand's commitment to high-tech, customer-centric service.
- The result is a more organized scheduling system that directly contributes to higher conversion rates from test drives to vehicle sales.



6. User Creation

- User Creation section to your report is essential to demonstrate how security and access are managed within WhatsNext Vision Motors.
- This section details the process of onboarding team members and assigning them the appropriate levels of authority.

a. New Vehicle Customer:

- The new data must be entered by a customer for their required actions in the application.
- Every field must be filled for the sake of processing the allocation of dealer to the customer.

The screenshot displays the 'New Vehicle Customer' form within the WhatsNext Vision Motors application. The form is a modal window with a title bar and a close button. It contains the following fields and controls:

- Vehicle Customer Name:** A text input field with a red asterisk indicating it is required.
- Email:** A text input field.
- Phone:** A text input field.
- Address:** A text input field.
- Preferred Vehicle Type:** A dropdown menu with 'None' selected.
- Owner:** A field showing 'Soma Shekar Rayavarapu' with a user icon.
- Buttons:** 'Cancel', 'Save & New', and 'Save' at the bottom right.

The background shows the 'Vehicle Customers' dashboard with a 'Recently Viewed' section containing two items: 'MSD' and 'Papa'. The dashboard has a search bar and various navigation tabs at the top.

b. New Vehicle Dealer:

- Vehicle dealer form is a form used to create a new vehicle dealer record, where the Vehicle Dealer Name is a required field.
- It allows users to enter important details such as dealer location, dealer code, phone number, and email address.
- The owner field is auto-assigned to the logged-in user, ensuring record

accountability.

https://orgfarm-2974de10-dev-ed.lightning.force.com/lightning/o/Vehicle_Dealer_c/new?count=3&nooverride=1&useRecordTyp...

Search...

New Vehicle Dealer

* = Required Information

Information

* Vehicle Dealer Name

Owner  Soma Shekar Rayavarapu

Dealer_Location

Dealer Code

Phone

Email

WhatNext Vision M... Vehicle Customers Vehicle Dealers Vehicle Orders Vehicle Service Requests Vehicle Test Drives Vehicles More

Vehicle Dealers

Recently Viewed 

2 items • Updated 4 minutes ago

Search this list...

	Vehicle Dealer Name	
1	☐ Zoro	
2	☐ Luffy	

c. New Vehicle Order

- It is a form used to create a new Vehicle Order record, where the Vehicle Order Number is system-generated.
- It allows users to select a Vehicle Customer and a Vehicle using searchable lookup fields.
- The form includes fields for order date and status to track the progress of the order.
- An assigned dealer can be selected, and action buttons like Save, Save & New, and Cancel are provided to complete the process.

The image displays two screenshots of the Salesforce Lightning interface for managing vehicle orders.

Top Screenshot: New Vehicle Order Form

The form is titled "New Vehicle Order" and includes a search bar at the top. The form fields are as follows:

- Vehicle Order Name:** A text field.
- Owner:** A dropdown menu showing "Soma Shekar Rayavarapu".
- Vehicle Customer:** A searchable lookup field with the placeholder text "Search Vehicle Customers..." and a magnifying glass icon.
- Vehicle:** A searchable lookup field with the placeholder text "Search Vehicles..." and a magnifying glass icon.
- Order Date:** A date picker field.
- Status:** A dropdown menu with "--None--" selected.

At the bottom of the form, there are three buttons: "Cancel", "Save & New", and "Save". A red asterisk indicates required information.

Bottom Screenshot: Vehicle Orders List

The screenshot shows the "Vehicle Orders" list in the Salesforce Lightning interface. The list is titled "Vehicle Orders" and includes a search bar and a "Recently Viewed" section. The list contains three items:

	Vehicle Order Name	
1	O-0004	
2	O-0003	
3	O-0002	

The list also includes a "Search this list..." bar and a set of action buttons (New, Import, Change Owner, Assign Label) at the top right.

d. New Vehicle Test Drive

- It is a form used to create a new Vehicle Test Drive record, where the Vehicle Test Name is a required field.
- It allows users to select a vehicle customer and a vehicle using searchable lookup fields.
- The form includes a test drive date and status to track the progress of the test drive.
- The owner is automatically assigned, and options like Save, Save & New, and Cancel are provided to manage the record.

The screenshot shows a 'New Vehicle Test Drive' form overlaying a Salesforce Lightning page. The form has a title bar 'New Vehicle Test Drive' and a sub-header 'Information'. It contains several fields: 'Vehicle Test Drive Name' (a required text field), 'Owner' (a dropdown menu showing 'Soma Shekar Rayavarapu'), 'Vehicle Customer' (a searchable lookup field), 'Vehicle' (a searchable lookup field), 'Test Drive Date' (a date picker), and 'Status' (a dropdown menu showing '--None--'). At the bottom of the form are three buttons: 'Cancel', 'Save & New', and 'Save'. The background page shows a 'Vehicle Test Drives' section with a 'Recently Viewed' list containing two items: 'rs' and 'rip'.

The screenshot shows the 'Vehicle Test Drives' list view in a Salesforce Lightning interface. The page has a header with navigation tabs: 'Vehicle Customers', 'Vehicle Dealers', 'Vehicle Orders', 'Vehicle Service Requests', 'Vehicle Test Drives' (selected), and 'Vehicles'. The 'Vehicle Test Drives' section has a 'Recently Viewed' dropdown and a search bar. Below the search bar is a table with two columns: 'Vehicle Test Drive Name' and a dropdown menu. The table contains two rows: '1 rs' and '2 rip'. At the bottom of the page is a URL bar showing the page's address.

e. New Vehicle Form

- It is a form used to create a new Vehicle record, where the Vehicle Name is a required field.
- It allows users to select the vehicle model and enter details such as stock quantity and price.
- The form includes a lookup to assign a vehicle dealer and a status to track availability.

- The owner is auto-assigned, and users can finalize the record using Save, Save & New, or Cancel options.

https://orgfarm-2974de2e10-dev-ed.develop.lightning.force.com/lightning/o/Vehicle__c/new?count=6&nooverride=1&useRecordTypeCheck...

WhatNext Vision

Vehicles

Recently Viewed

2 items • Updated a few seconds ago

Vehicle Name

1 Skoda

2 BMW

New Vehicle

* = Required Information

Information

*Vehicle Name

Owner

Soma Shekar Rayavarapu

Vehicle Model

--None--

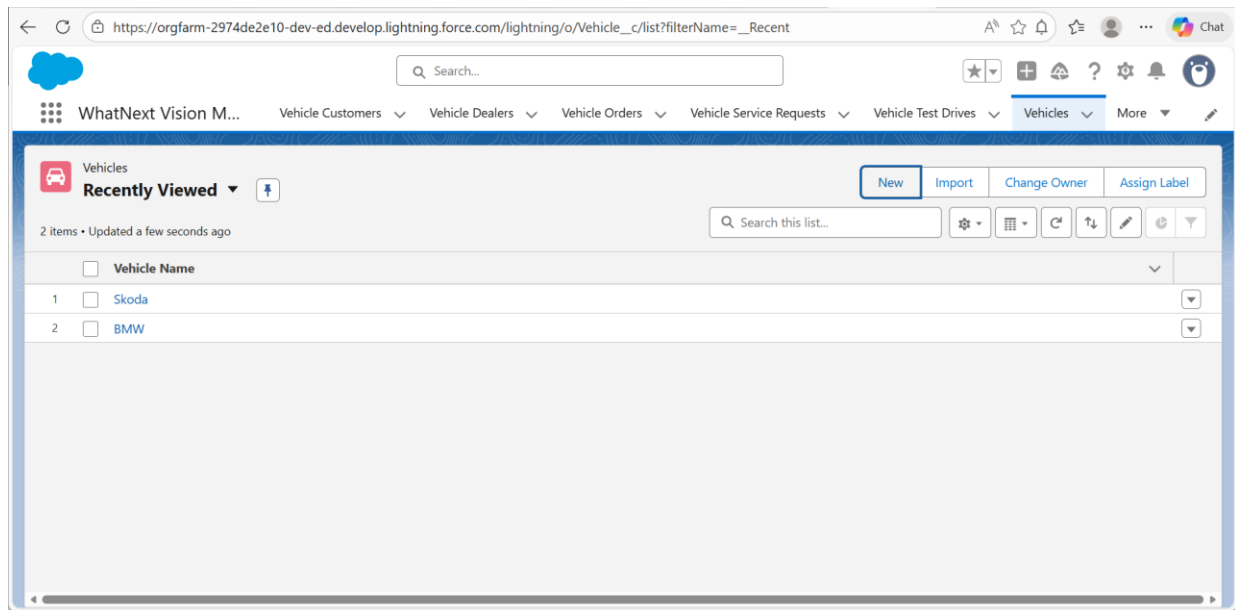
Stock Quantity

Price

Vehicle Dealer

Search Vehicle Dealers...

Cancel Save & New Save



7. Email Template and Alerts

- This email is a reminder generated automatically through a Salesforce Flow execution.
- The flow is triggered based on predefined conditions, such as a scheduled vehicle test drive date approaching.
- Once the criteria are met, the flow sends an automated notification email to the customer.
- The purpose of this reminder is to inform the user that their test drive is scheduled for the next day.
- It helps improve customer communication and reduces the chances of missed appointments.

- The email includes essential details like the test drive reference ID for identification.
- A support email address is provided in case the customer needs to reschedule or ask for assistance.
- Since the email is system-generated, Gmail may sometimes classify it as spam.
- This can happen if similar automated messages were previously marked as spam.
- Overall, the flow ensures timely reminders and enhances the efficiency of the test drive management process.

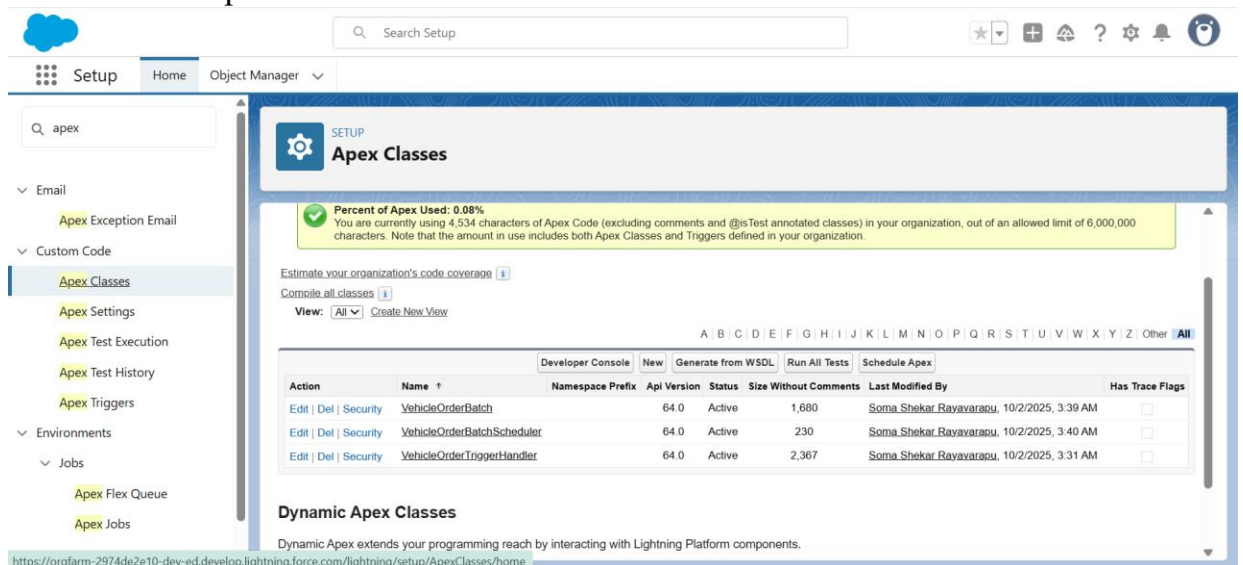
8. APEX Triggers

1. Trigger 1: VehicleOrderTrigger (Before Insert / Before Update)

- This trigger validates vehicle availability before an order is saved into the system.
- It prevents users from placing or updating an order when the vehicle stock quantity is zero.
- By using addError, it ensures data integrity and avoids invalid orders at the database level.

2. Trigger 2: VehicleOrderTrigger (After Insert / After Update)

- This trigger executes after the order is successfully saved in Salesforce.
- It automatically reduces the vehicle stock quantity when an order status is marked as Confirmed.
- It keeps vehicle inventory synchronized with confirmed orders, ensuring real-time stock updates.



Search Setup

Setup Home Object Manager

Q apex

Email

- Apex Exception Email

Custom Code

- Apex Classes
- Apex Settings
- Apex Test Execution
- Apex Test History
- Apex Triggers

Environments

- Jobs
- Apex Flex Queue
- Apex Jobs

Apex Classes

Percent of Apex Used: 0.08%
You are currently using 4,534 characters of Apex Code (excluding comments and @isTest annotated classes) in your organization, out of an allowed limit of 6,000,000 characters. Note that the amount in use includes both Apex Classes and Triggers defined in your organization.

Estimate your organization's code coverage

Compile all classes

View: All Create New View

		Developer Console	New	Generate from WSDL	Run All Tests	Schedule Apex		
Action	Name	Namespace Prefix	Api Version	Status	Size Without Comments	Last Modified By	Has Trace Flags	
Edit Del Security	VehicleOrderBatch		64.0	Active	1,680	Soma Shekar Rayavarapu, 10/2/2025, 3:39 AM		
Edit Del Security	VehicleOrderBatchScheduler		64.0	Active	230	Soma Shekar Rayavarapu, 10/2/2025, 3:40 AM		
Edit Del Security	VehicleOrderTriggerHandler		64.0	Active	2,367	Soma Shekar Rayavarapu, 10/2/2025, 3:31 AM		

Dynamic Apex Classes

Dynamic Apex extends your programming reach by interacting with Lightning Platform components.

<https://orgfarm-2974de2e10-dev-ed.develop.lightning.force.com/lightning/setup/ApexClasses/home>

Process of Creation:

Creation of Objects:

Setup

Home

Object Manager

Search Setup

Star

Plus

Shield

Help

Settings

Notifications

User

SETUP

Object Manager

6 Items, Sorted by Label

veh

Schema Builder

Create

1. Vehicle

https://orgfarm-1480109947-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lg5000000XY9/Details/view

Star

Plus

Shield

Help

Settings

Notifications

User

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Vehicle

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Details

Description

API Name

Vehicle__c

Custom

✓

Singular Label

Vehicle

Plural Label

Vehicles

Enable Reports

✓

Track Activities

Track Field History

Deployment Status

Deployed

Help Settings

Standard salesforce.com Help Window

Edit

Delete

2. Vehicle Customer

The screenshot shows the Salesforce Setup interface for the 'Vehicle Customer' object. The left sidebar contains a 'Details' menu with options: Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, and Scoping Rules. The main content area is titled 'Vehicle Customer' and contains a 'Details' section with the following fields:

Field	Value
Description	
API Name	Vehicle_Customer_c
Custom	✓
Singular Label	Vehicle Customer
Plural Label	Vehicle Customers
Enable Reports	✓
Track Activities	
Track Field History	
Deployment Status	Deployed
Help Settings	Standard salesforce.com Help Window

Buttons for 'Edit' and 'Delete' are located in the top right corner of the details section.

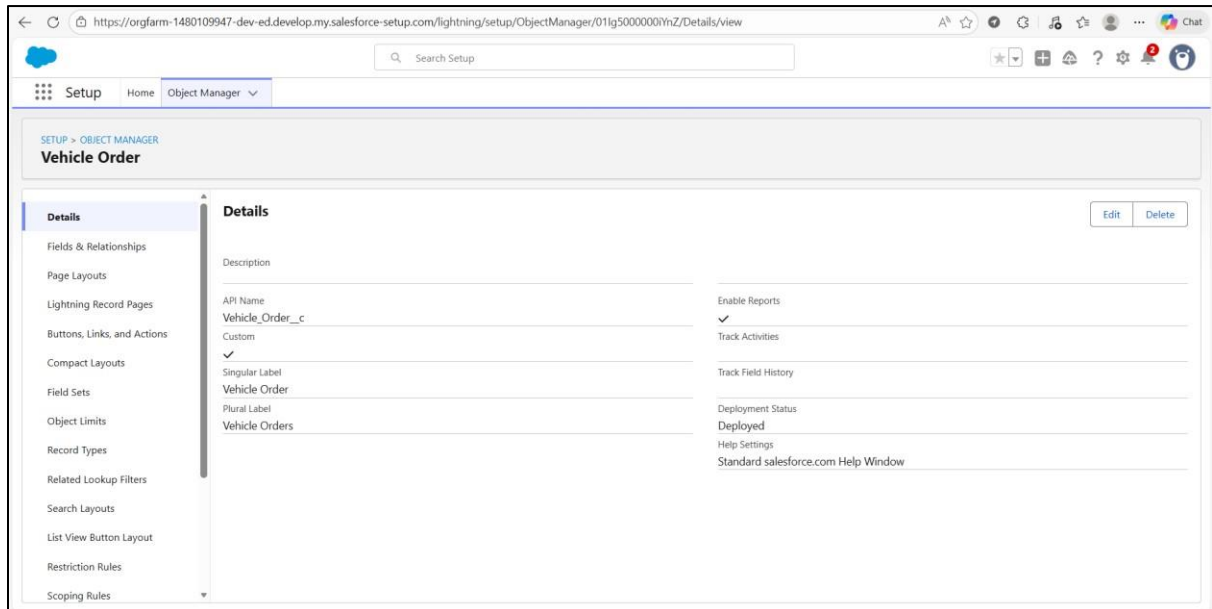
3. Vehicle Dealer

The screenshot shows the Salesforce Setup interface for the 'Vehicle Dealer' object. The left sidebar contains a 'Details' menu with options: Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, and Scoping Rules. The main content area is titled 'Vehicle Dealer' and contains a 'Details' section with the following fields:

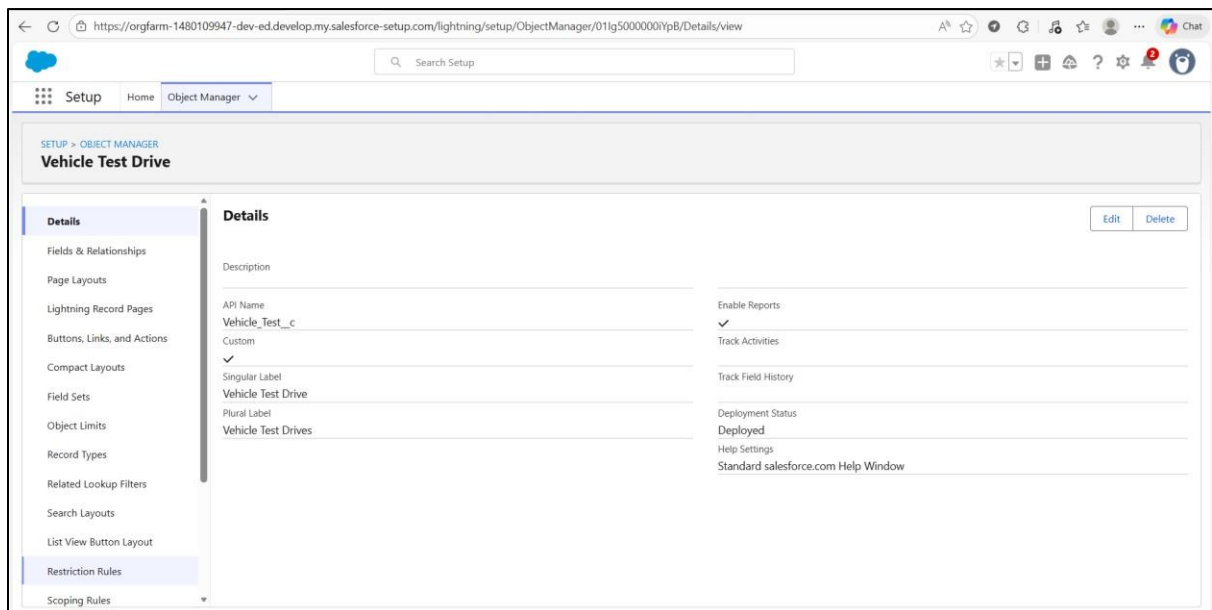
Field	Value
Description	
API Name	Vehicle_Dealer_c
Custom	✓
Singular Label	Vehicle Dealer
Plural Label	Vehicle Dealers
Enable Reports	✓
Track Activities	
Track Field History	
Deployment Status	Deployed
Help Settings	Standard salesforce.com Help Window

Buttons for 'Edit' and 'Delete' are located in the top right corner of the details section.

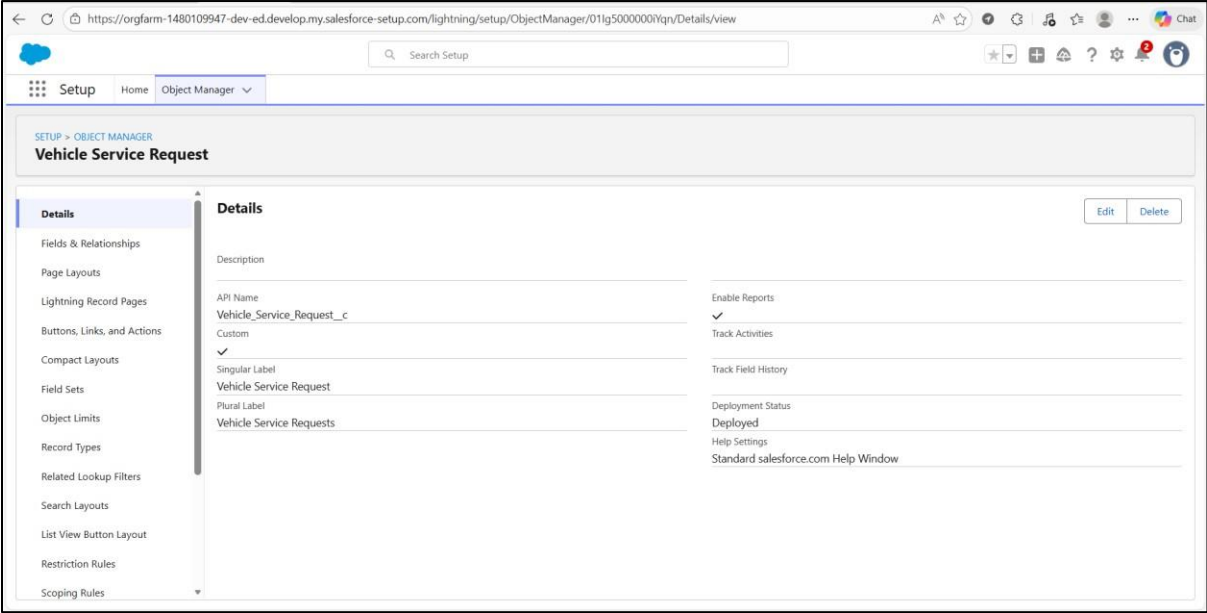
4. Vehicle Order



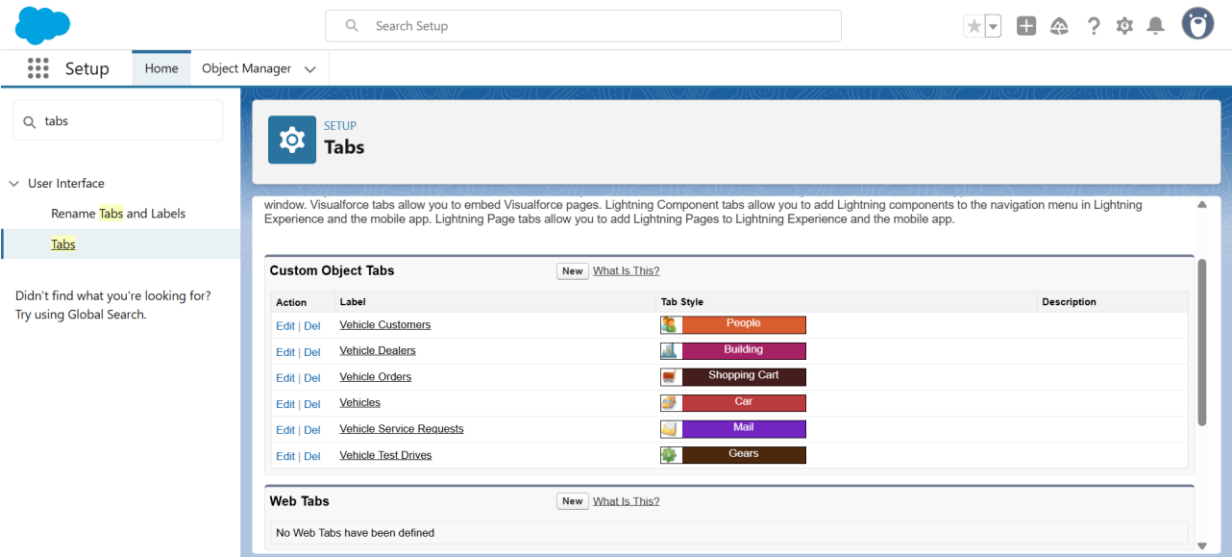
5. Vehicle Test Drive



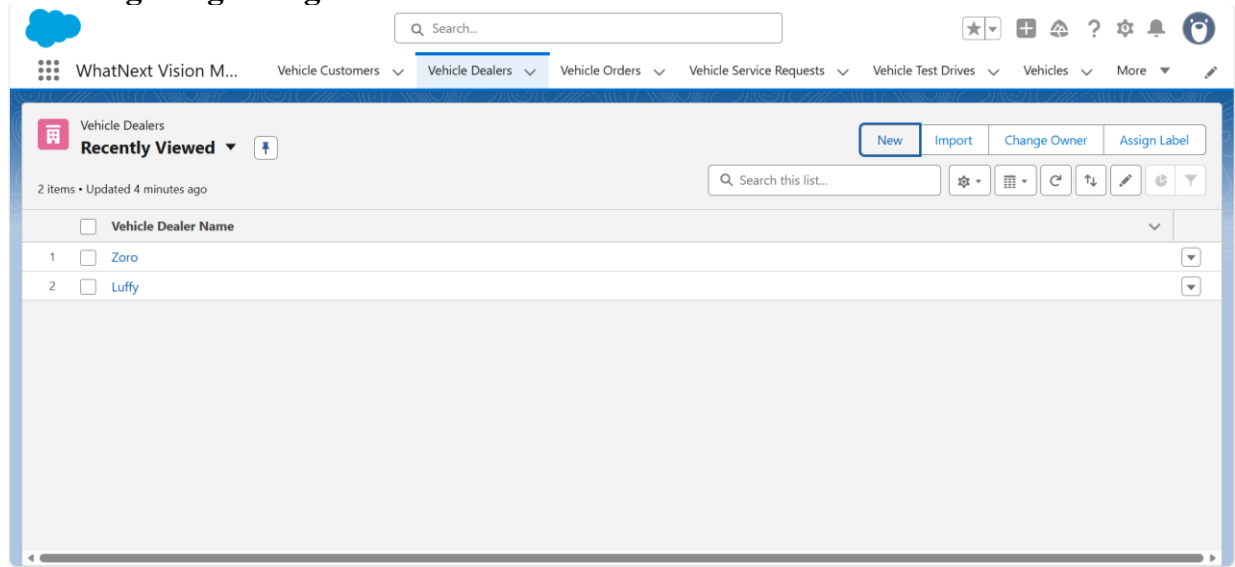
6. Vehicle Service Request



Creation of Tabs

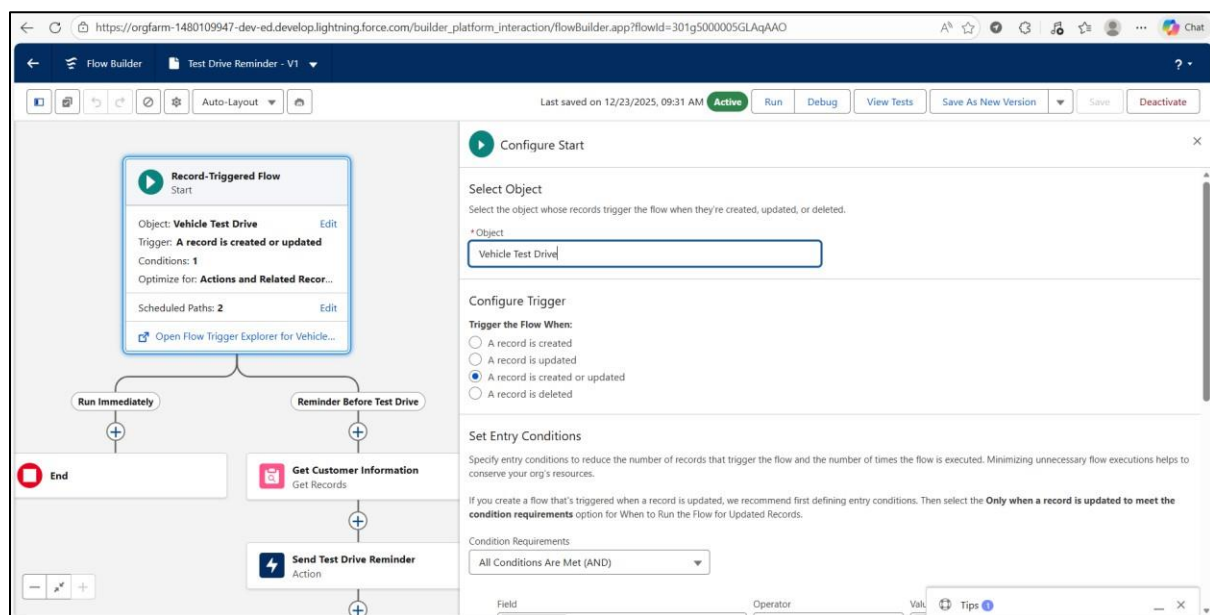


Creating a Lightning APP

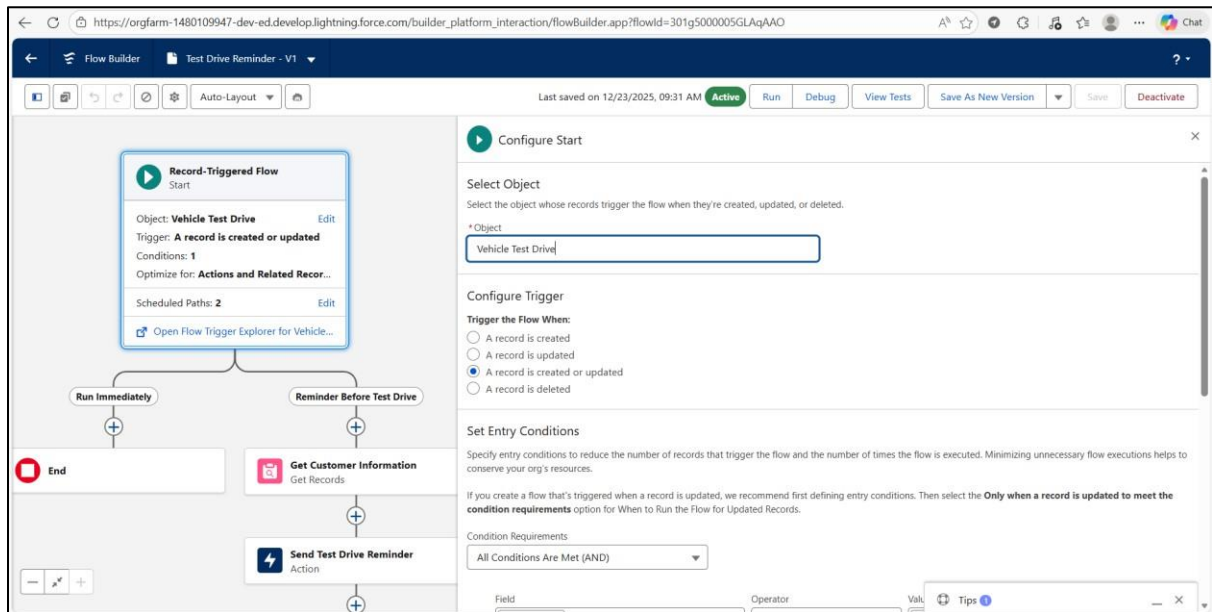


Automation

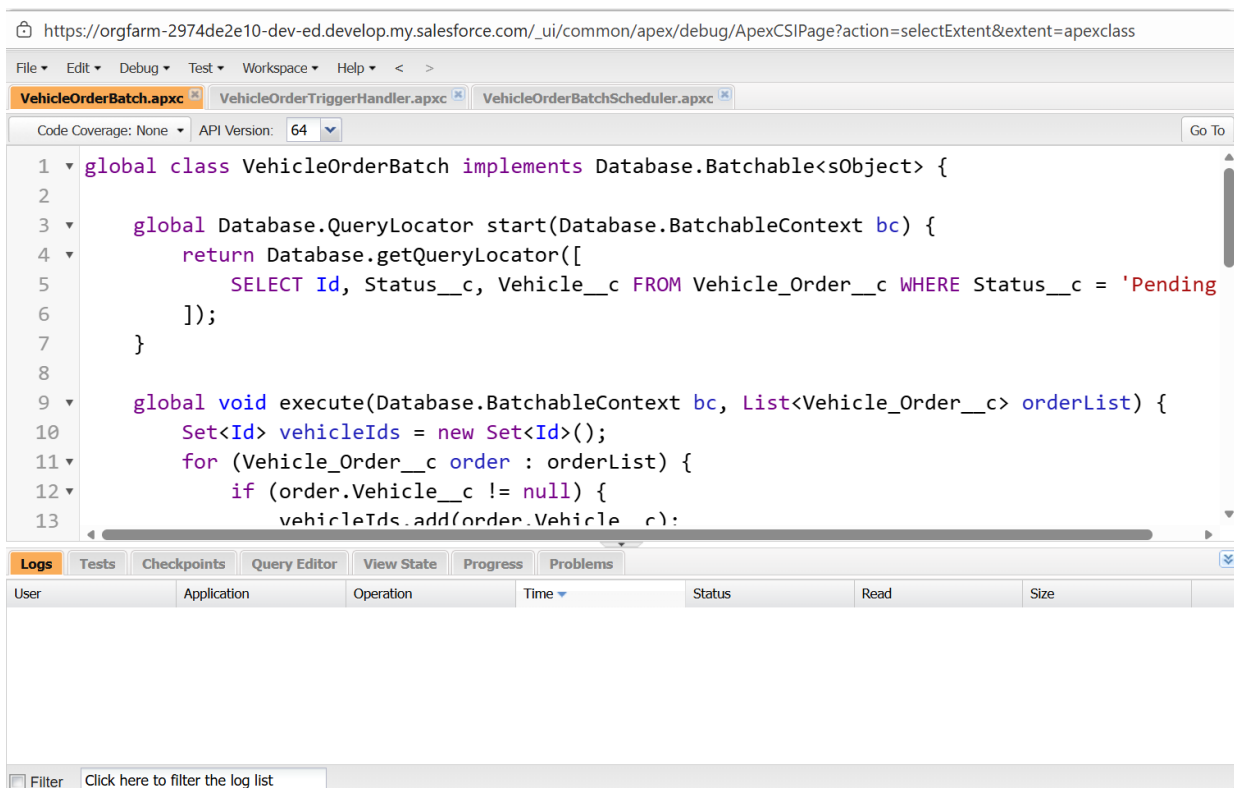
- Test Drive Reminder Flow



- **Auto Assign Dealer Flow**



Apex and Trigger Batch Jobs



Summary

WhatsNext Vision Motors, a leader in the automotive industry, is driving transformation in the mobility sector through innovative technology and customer-focused solutions. To strengthen its commitment to excellence, the company has launched an ambitious Salesforce project designed to enhance customer experiences while streamlining operational processes.

At the core of this initiative is the improvement of the customer ordering journey. The system has been built to automatically suggest the nearest dealer location based on a customer's address. This feature greatly increases convenience, reduces effort, and ensures that customers can quickly connect with the most accessible dealership when placing an order.

Another critical aspect of the project addresses the challenge of stock availability, a common issue in the automotive sector. The system prevents customers from placing orders for vehicles that are out of stock, ensuring that only available models can be purchased. This proactive measure eliminates confusion, avoids disappointment, and strengthens trust by providing a transparent and reliable ordering process.

To further improve efficiency, the project incorporates a scheduled process for updating the status of bulk order records. Orders are automatically updated depending on stock availability: if a vehicle is unavailable, the status is set to "Pending," while available vehicles are marked as "Confirmed." This automation ensures accurate reflection of order statuses and provides customers with clear communication about their purchase progress.

The implementation of this Salesforce project is expected to deliver significant benefits. By streamlining the ordering process and ensuring accurate stock validation, WhatsNext Vision Motors can enhance customer satisfaction and loyalty. At the same time, automation reduces the administrative burden on staff, allowing employees to focus on strategic tasks that require human expertise. This not only boosts productivity but also enables the company to respond more quickly to

market demands and evolving customer needs.

